

Long Quiz III

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Instructions. Closed book, closed notes. Total: **100 points**. Time: **120 minutes**. Write your name and ID at the top of every page. Each answer requires **at most two sentences**; use only the space provided. Anything outside the allotted space is considered rough work and **will not be graded**. Problem names are given; **definitions are not** — you are expected to know them. Complete proofs are **not** required; a concise one-sentence argument suffices for each question. **IDK Rule:** Writing “I don’t know” (IDK) for any question earns **25%** of that question’s marks.

NP-Hardness

Problem 1. [4 pts]State the decision version of **Problem A**.

Problem 2. [4 pts]Prove that **Problem A** is in NP.

Problem 3. [4 pts]State the decision version of **Problem B**.

Problem 4. [4 pts]Prove that **Problem B** is NP-hard by giving a polynomial-time reduction from **Problem A**, and prove correctness.

Problem 5. [4 pts]Give a polynomial-time reduction from **Problem B** to **Problem A**, and prove correctness.

Problem 6. [4 pts]

Give a polynomial-time reduction from **Problem A** to **Problem C**, and prove correctness in one sentence.

Problem 7. [4 pts]

Give a polynomial-time reduction from **Problem B** to **Problem C**, and prove correctness in one sentence.

Problem 8. [4 pts]

State the recurrence for the Dynamic Programming solution to **Problem X**.

Problem 9. [4 pts]

Prove that **Problem D** is NP-hard by giving a polynomial-time reduction from **Problem E**.

Problem 10. [4 pts]

Prove that **Problem F** is in NP.

Problem 11. [4 pts]

Prove that **Problem F** is NP-hard by giving a polynomial-time reduction from **Problem G**.

Problem 12. [4 pts]

Prove that **Problem H** is in NP.

Problem 13. [4 pts]

Prove that **Problem I** is NP-hard by reducing from **Problem H**.

Problem 14. [4 pts]

Prove that **Problem H** is NP-hard by reducing from **Problem I**.

Problem 15. [4 pts]

Prove that **Problem J** is NP-hard by reducing from **Problem I**.

Problem 16. [4 pts]

Prove that **Problem K** is NP-hard by reducing from **Problem H**.

This problem is about network flow.

Problem 22. [4 pts]

Consider a greedy approximation algorithm for a **Problem P**. State lower bound(s) on OPT.

Problem 23. [4 pts]

Using the bounds from the previous question, prove the approximation ratio of the algorithm.

Problem 24. [4 pts]

Give a one-sentence family of instances that achieves the approximation ratio.

Problem 25. [4 pts]

Give a polynomial-time algorithm for **Problem Q**.
